

1 Part 3: Introduction to Modelling

2 A fundamental step in many data analyses is to **model** raw, observable
3 data, i.e. to replace the complex processes that generate these data with a
4 mathematically and computationally simpler fiction.

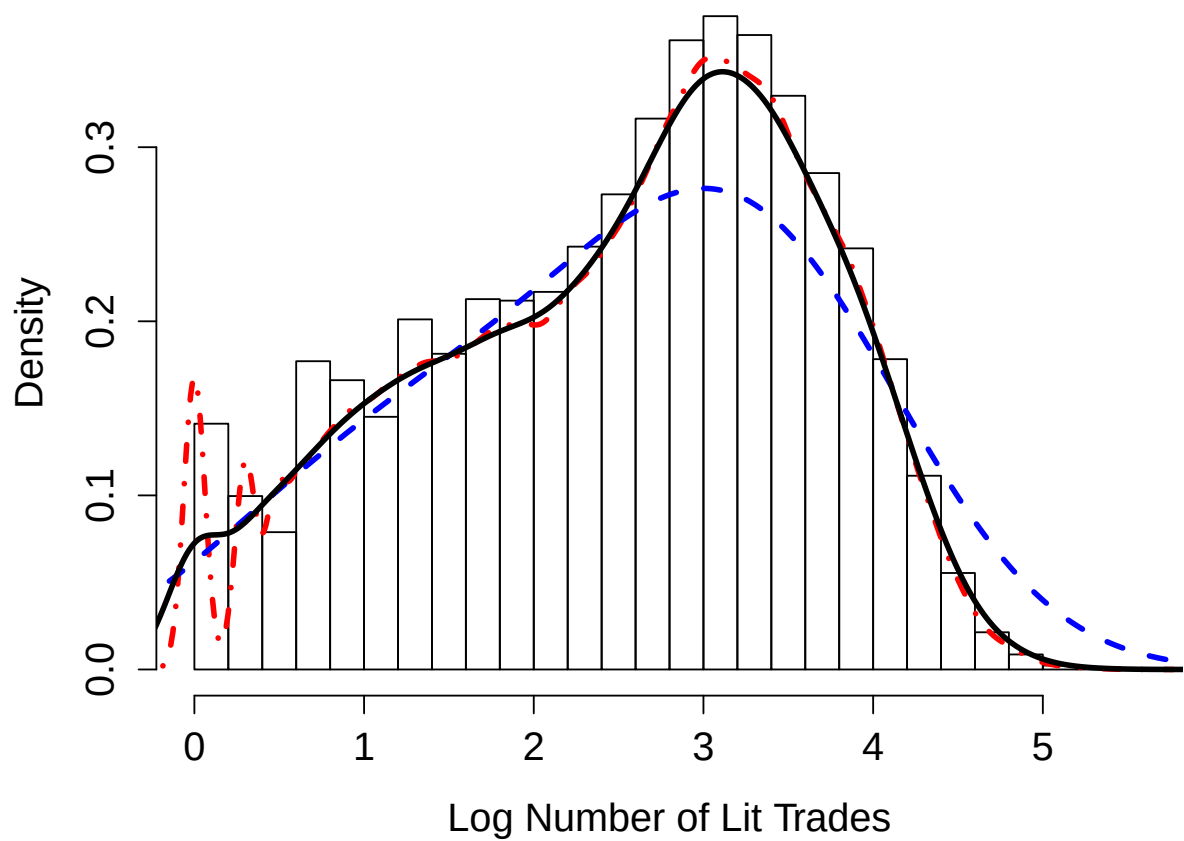
5 You have likely heard the expression, "**All models are wrong; some are**
6 **useful.**"¹ That is a good mentality to take towards the modelling process:
7 We do not believe that the data were actually generated by, for example,
8 drawing from a normal distribution. Instead, it **may** be the case that the
9 normal provides a sufficiently accurate approximation to that process.

¹For a history of this saying, see https://en.wikipedia.org/wiki/All_models_are_wrong.

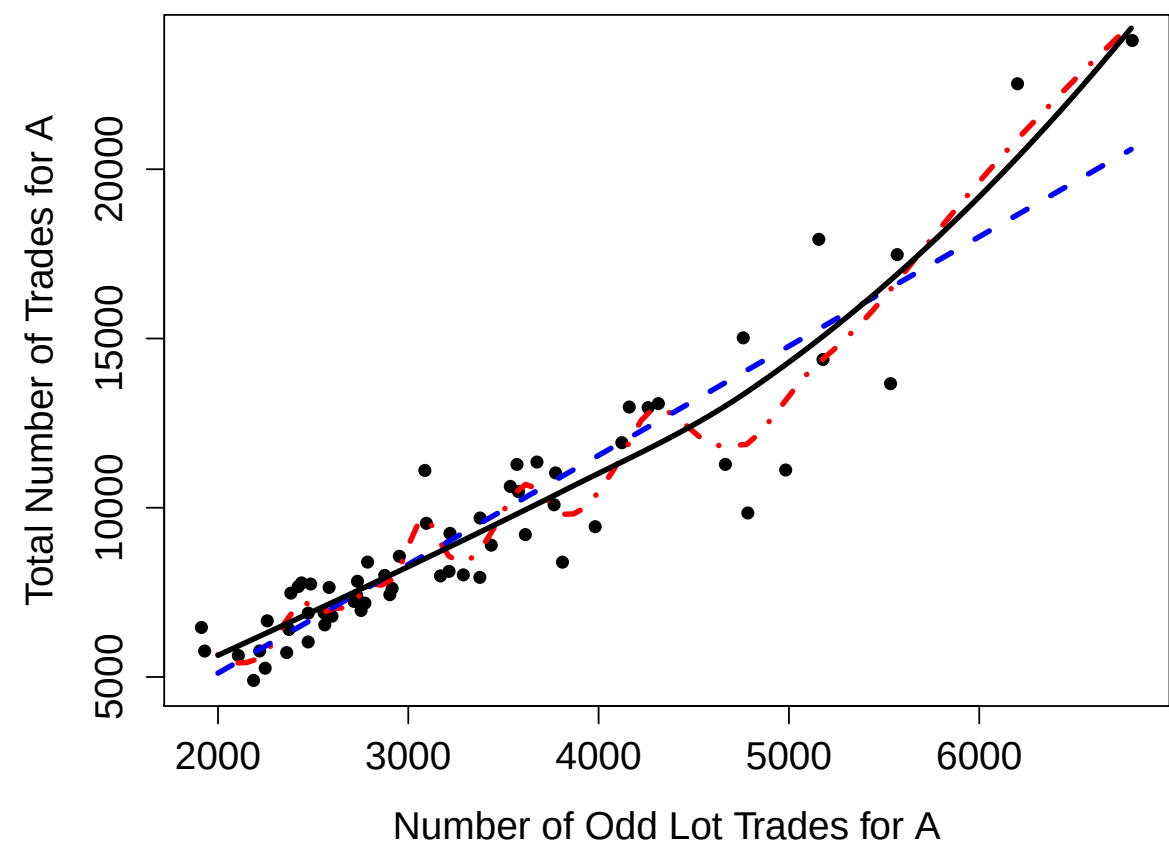
1 A key heuristic in the statistical modelling process is that of **smoothness**.

2 We believe that most relationships, and most distributions, are fundamen-
3 tally **smooth**, i.e., we don't try to fit to every little fluctuation that our sam-
4 ple may possess.

5 Of course, this idea can be taken too far, and **oversmoothing** can lead to
6 missing important features that are present in data.



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- 1 **Exercise:** Explain how the three proposed models on the previous slides
2 seem to perform, in regards to the appropriate amount of smoothing of the
3 histogram and scatter plot.

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- 1 **Exercise:** Explain why overfitting is such a significant concern.

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1 In many situations, there are one or more unknown **population quantities**
2 that are of interest to estimate. For example,

- 3 • The probability that the return exceeds c tomorrow
- 4 • The “beta” for some stock over the past year.
- 5 • The volatility for some stock over the past three months.
- 6 • The mean of some distribution of interest.
- 7 • The variance of some distribution of interest.
- 8 • The λ parameter from some Exponential distribution.

9 The model serves as a means to estimating such a quantity, since we imag-
10 ine that the true model, if known, would provide the true value of the
11 unknown. An estimated model provides an estimate of the unknown.

1 From a statistical standpoint, we say that a probability model is **incom-**
2 **pletely specified**. I.e., a particular model is assumed, but there are still
3 unknown components, often real-valued **parameters**.

4 We use available data (a **sample**) to estimate these parameters.

5 The quantities of interest such as those listed above are either the parame-
6 ters themselves, or some function of those parameters.

7 **Example:** In a certain situation, an individual is comfortable assuming that
8 $X_1, X_2, \dots, X_n$ are **independent and identically distributed (i.i.d.)** with the
9 Normal distribution with mean μ and variance σ^2 . These parameters are
10 unknown, and must be estimated from data.

1 **Exercise:** Suppose someone wants to estimate the probability that tomor-
2 row's return will be less than c . Describe steps as to how a model could be
3 constructed and the quantity of interest estimated.

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1 Terminology

2 The **parameters** are the constants that distinguish probability models in the
3 same "family." Examples of "families" include "normal," "exponential,"
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**.

8 An **estimator** is a statistic which is used to estimate an unknown parameter
9 or an unknown function of a parameter. It is also a **random variable**.

10 The "hat notation" is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

- 1 **Exercise:** Explain the sense in which an estimator is appropriately thought
2 of as a random variable.

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1 Performance of Estimators

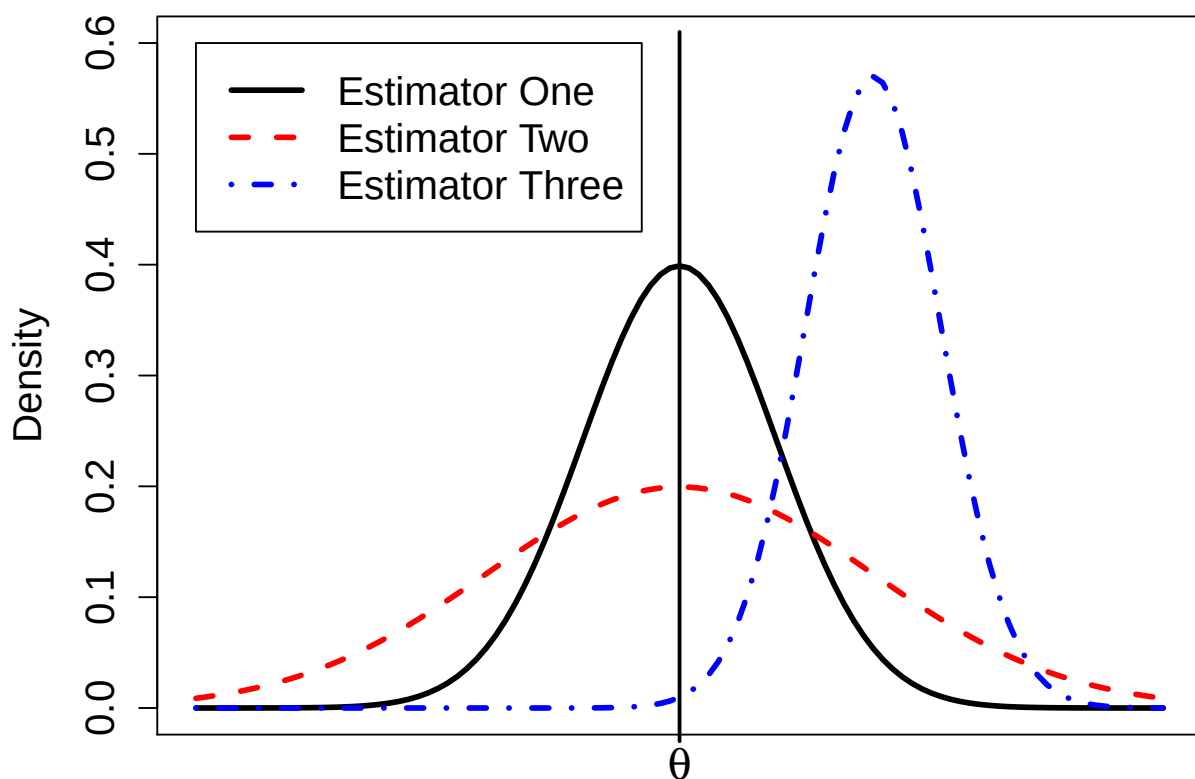
2 Our objective is to estimate unknowns, i.e., to construct estimators. For
3 any situation, there are many choices of estimator, so we require methods
4 to assess the performance of an estimator.

5 The “performance” is closely tied to the distribution of the estimator. The
6 distributions we learned in Probability (plus some more) are useful for this
7 purpose.

8 An ideal estimator $\hat{\theta}$ for θ would have two properties:

- 9 1. The estimator should be **accurate**, i.e., its distribution would be cen-
10 tered on the true value θ .
- 11 2. The estimator should be **precise**, i.e., its distribution has a small spread
12 (small variance)

- 1 **Example:** The figure on the following page compares the distributions of
2 three hypothetical estimators for a parameter θ , call them $\hat{\theta}_1$, $\hat{\theta}_2$, and $\hat{\theta}_3$.
3 We note that $E(\hat{\theta}_1) = \theta$, so it is optimally accurate. Also, $E(\hat{\theta}_2) = \theta$.
4 But, $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, so $\hat{\theta}_1$ is more precise than $\hat{\theta}_2$.
5 So, we would prefer $\hat{\theta}_1$ to $\hat{\theta}_2$.
6 Also, $V(\hat{\theta}_3) < V(\hat{\theta}_1)$, but $\hat{\theta}_3$ has terrible accuracy.



- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.
- 7 An estimator $\hat{\theta}$ is said to be the **minimum variance unbiased estimator for**
- 8 **θ (MVUE)** if $\hat{\theta}$ is unbiased for θ and for any other unbiased estimator $\hat{\theta}'$,
- 9 $V(\hat{\theta}) \leq V(\hat{\theta}')$.

- 1 **Exercise:** Suppose that X has the binomial(n, p) distribution. It is com-
- 2 mon to estimate p using $\hat{p} = X/n$, the sample proportion. Is this estimator
- 3 unbiased? What is its variance and standard error?

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1 There is often a “tradeoff” between bias and variance (accuracy and preci-
2 sion). This is called the **bias-variance tradeoff**.

3 In some cases, allowing some bias leads to reduced variance.

4 This tradeoff is made formal via the **mean squared error (MSE)**. For an
5 estimator $\hat{\theta}$ for θ ,

6
$$\text{MSE}_{\theta}(\hat{\theta}) = E\left((\hat{\theta} - \theta)^2\right)$$

7 i.e., it is the expected squared error in estimating θ when using $\hat{\theta}$.

8 One can show that

9
$$\text{MSE}_{\theta}(\hat{\theta}) = V(\hat{\theta}) + \text{bias}_{\theta}^2(\hat{\theta})$$

10 where

11
$$\text{bias}_{\theta}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

1 **Exercise:** How does the bias/variance tradeoff relate to our initial discus-
2 sion of smoothness?

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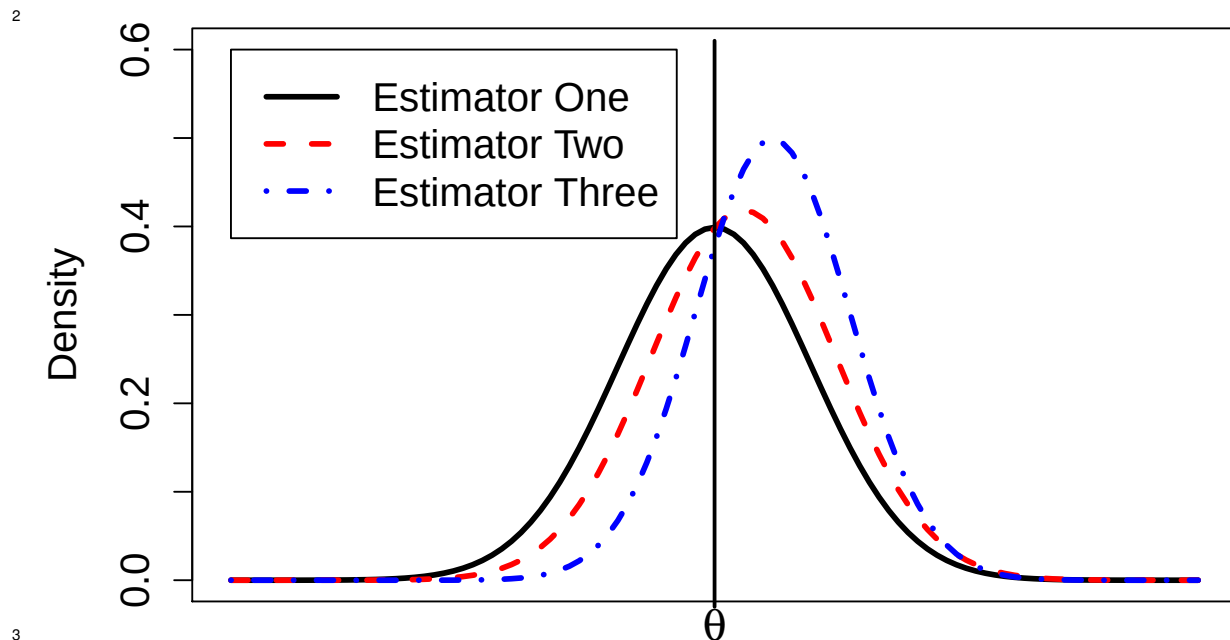
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- 1 In the example below, all three estimators have the same MSE.



1 Important Cases

- 2 There are a few estimation situations that worth special attention.

- 3 **Case 1:** The first we have already explored, that of estimating a population
 4 proportion. We found that the sample proportion is unbiased and has a
 5 standard error of

6
$$\sqrt{\frac{p(1-p)}{n}}$$

- 7 Of course, we do not know the value of p , so typically it is replaced by its
 8 best estimate and the approximate SE is reported as

9
$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

1 **Exercise:** We have derived the mean and variance of the sample propor-
2 tion, but what is its distribution?

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1 **Case 2:** Suppose that $X_1, X_2, \dots, X_n$ are random variables and $E(X_i) = \mu$
2 and $V(X_i) = \sigma^2$ for all i . Further assume that these random variables
3 are uncorrelated (meaning that every pair of random variables is uncorre-
4 lated).

5 Then, the standard estimator for μ is the sample mean, $\bar{X}$, and the standard
6 estimator for σ^2 is the sample variance,

7
$$s^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$$

8 Note that the case where $X_1, X_2, \dots, X_n$ are i.i.d. is a special instance of
9 this situation.

- 1 **Exercise:** Derive the mean, variance, and standard error of the sample
2 mean. What is its MSE as an estimator of μ ?

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- 1 **Exercise:** If we did assume that $X_1, X_2, \dots, X_n$ are i.i.d., what could we say
2 about the distribution of the sample mean?

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- 1 **Exercise:** As defined, the sample variance is an unbiased estimator of σ^2 .
2 (The proof is not difficult, but a little tedious.)

- 3 Consider the following question: What choice of a minimizes the quantity

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$$\sum_{i=1}^n (X_i - a)^2 ?$$

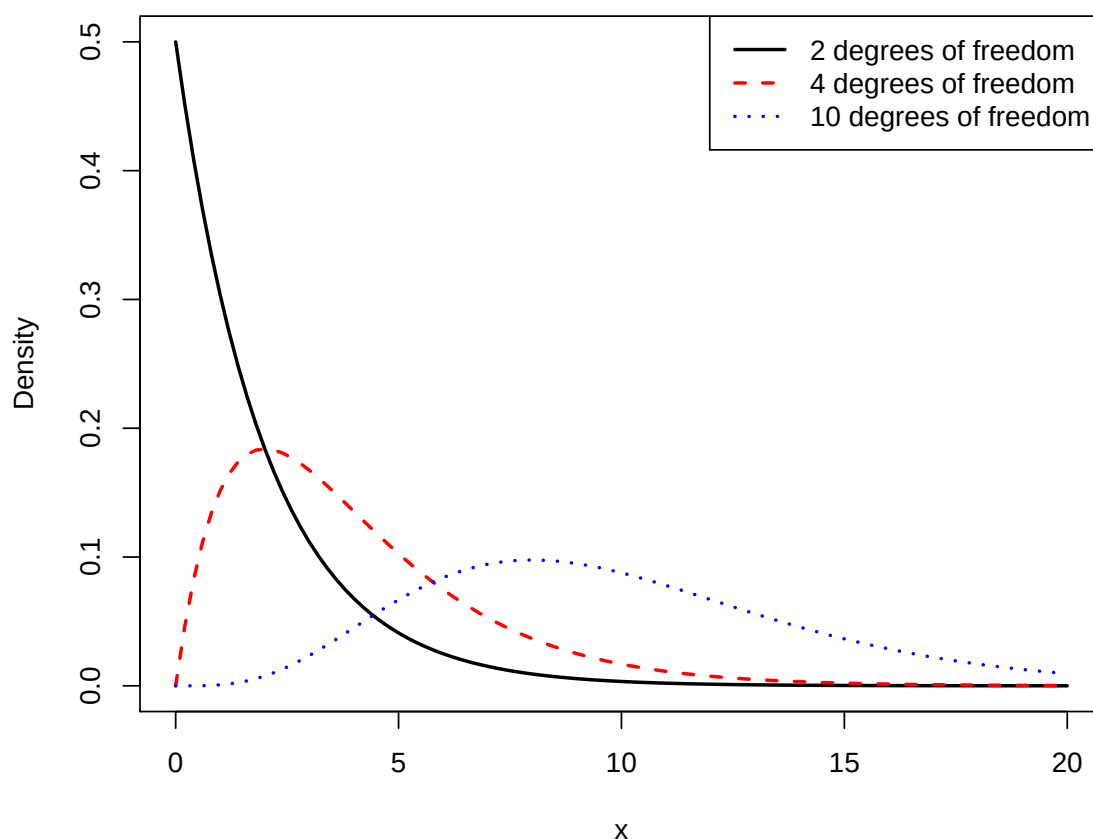
- 5 How does this result suggest it would be a good idea to estimate σ^2 by
6 dividing the sum of squares by a constant less than n ?

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- 1 **Case 3:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the normal distribution
2 with mean μ and variance σ^2 .
- 3 Of course, this is a special instance of “Case 2” presented above, and the
4 sample mean and sample variance remain unbiased estimators of μ and
5 σ^2 , respectively.
- 6 We can make some stronger statements, as well.
- 7 First, we know that $\bar{X}$ has **exactly** the normal distribution with mean μ
8 and variance σ^2/n for **any** sample size.
- 9 Further results depend on properties of probability distributions that we
10 will present now.

- 1 The **Chi-squared distribution with k degrees of freedom** is a special case
2 of the Gamma distribution. In particular, if X has the $\text{Gamma}(k/2, 1/2)$
3 distribution, then X has the Chi-square distribution with k degrees of free-
4 dom.
- 5 Also, if $Z_1, Z_2, \dots, Z_k$ are i.i.d. standard normal random variables, then
- $$6 \quad X = \sum_{i=1}^k Z_i^2$$
- 7 has the Chi-squared distribution with k degrees of freedom.
- 8 If X has this distribution, then $E(X) = k$ and $V(X) = 2k$.
- 9 Plots of pdfs from the Chi-squared distribution are shown on the following
10 slide.



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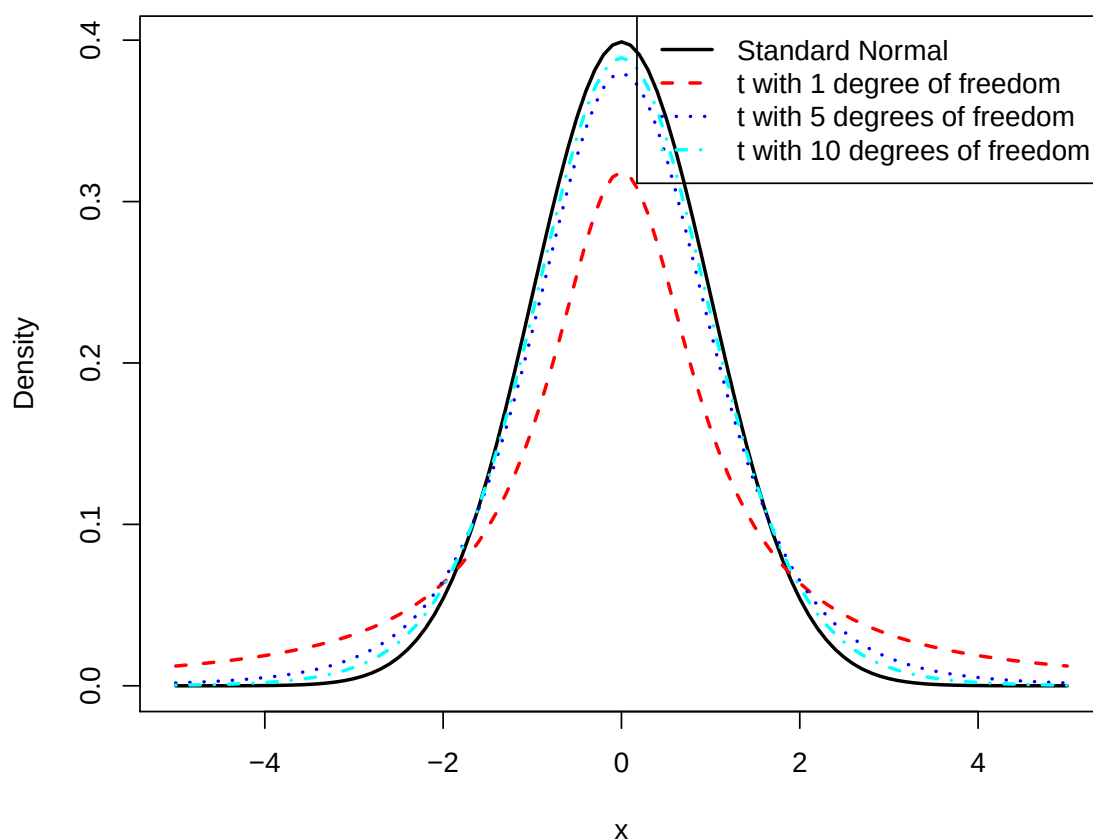
1 The *t*-distribution with ν degrees of freedom is defined to be a random
 2 variable T that can be constructed as

$$3 \quad T = \frac{Z}{\sqrt{X/\nu}}$$

4 where Z is a random variable with the standard normal distribution, and
 5 X is a random variable with the Chi-squared distribution with ν degrees
 6 of freedom. Further, Z and X are assumed independent.

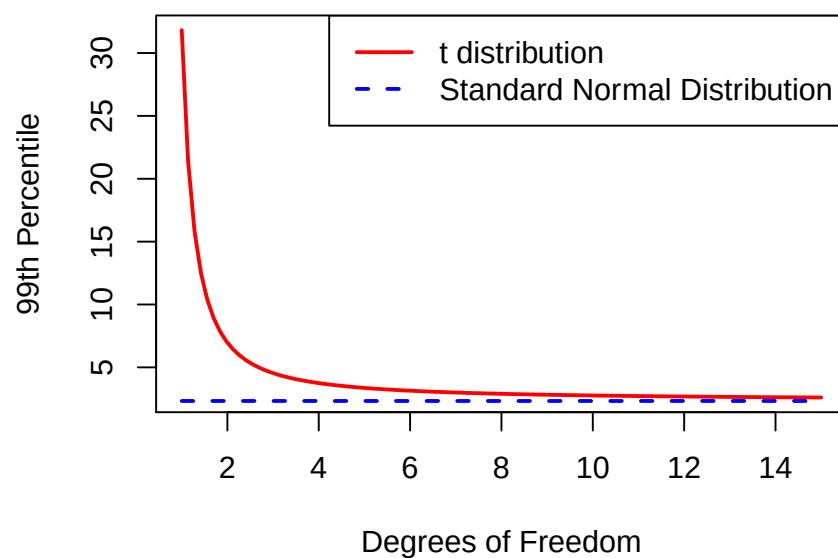
7 This distribution has a “normal-shaped” pdf, but has heavier tails than
 8 the normal distribution. As ν increases, the distribution converges to the
 9 standard normal distribution.

10 It holds that $E(T) = 0$ when $\nu \geq 2$ and that $V(T) = \nu/(\nu - 2)$ when $\nu \geq 3$.



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- 1 The figure below makes clearer how much additional probability is in the
 2 tails of the t -distribution compared to the standard normal. For $\nu = 1$,
 3 1% of the observations will be larger than 30. The 99th percentile of the
 4 standard normal distribution is 2.33.



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1 Now we can state the four **classic results** that hold under our “Case 3:”

2 Assuming that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, then

3 1. $\bar{X}$ and s^2 are independent.

4 2. The quantity

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

6 has the standard normal distribution.

7 3. The quantity

$$\frac{(n-1)s^2}{\sigma^2}$$

9 has the Chi-squared distribution with $n-1$ degrees of freedom.

10 4. The quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

12 has the t -distribution with $n-1$ degrees of freedom.

1 **Exercise:** Under “Case 3” it is easier to show that s^2 is an unbiased estima-
2 tor of σ^2 , and its also easy to derive the variance of this estimator. Do these
3 now.

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1 Confidence Intervals

2 The already-mentioned standard error (SE) is a useful tool for quantifying
3 error in an estimate, but a common alternative is the use of **confidence**
4 **intervals** for unknown parameters.

5 Formally, we call (L, U) a **$100(1 - \alpha)\%$ confidence interval for θ** if

$$6 \quad P(L \leq \theta \leq U) = 1 - \alpha$$

7 Hence, if want a “95% confidence interval,” set $\alpha = 0.05$. Other common
8 choices are 90% and 99% confidence intervals ($\alpha = 0.10$ and $\alpha = 0.01$, re-
9 spectively).

1 An appropriate statement regarding a confidence interval is, for example,
2 “I am 95% confident that the true value of θ is between 10.6 and 11.7.”

3 It is **not** appropriate to say that “The probability that θ is between 10.6
4 and 11.7 is 0.95.” This is incorrect because there is nothing random in that
5 statement: θ , 10.6 and 11.7 are all constants.

6 The randomness is in the procedure itself. If one constructs 95% confidence
7 intervals in a wide range of situations, then 95% of those intervals will
8 cover the true value of the parameter.

9 There is an alternative approach to statistical inference, **Bayesian inference**,
10 in which parameters are treated as random variables. More on that later.

- 1 **Exercise:** Use our “classic results” above for “Case 3” to derive confidence
2 intervals for μ and σ^2 in that situation. For estimating μ , onsider both the
3 instances where σ^2 is known and unknown.

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1 The next slides present four important confidence intervals that are com-
2 monly used.

3 *Confidence Interval for Population Mean, μ , when n is large ($n \geq 30$)*

4 Assume that the sample is i.i.d. from a distribution with mean μ and vari-
5 ance σ^2 , with both μ and σ^2 unknown. Then, a $100(1 - \alpha)\%$ confidence
6 interval for μ is

7
$$\bar{X} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

8 where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal
9 distribution. In R notation, z_a equals `qnorm(1-a)`.

1 *Confidence Interval for Population Mean, μ , when n is small ($n < 30$)*

2 Assume that the sample is i.i.d. from the normal distribution with mean
3 μ and variance σ^2 , with both μ and σ^2 unknown. Then, a $100(1 - \alpha)\%$
4 confidence interval for μ is

5
$$\bar{X} \pm t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

6 where $t_{a,v}$ is such that $P(T > t_{a,v}) = a$ when T has the t -distribution with
7 v degrees of freedom. In R notation, $t_{a,v}$ equals `qt(1-a, v)`.

1 *Confidence Interval for Population Variance, σ^2*

2 Assume that the sample is i.i.d. from the normal distribution with mean
 3 μ and variance σ^2 , with both μ and σ^2 unknown. Then, a $100(1 - \alpha)\%$
 4 confidence interval for σ^2 is

$$5 \left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{(1-\alpha/2), n-1}^2} \right)$$

6 where $\chi_{a,v}^2$ is such that $P(U > \chi_{a,v}^2) = a$ when U has the chi-squared
 7 distribution with v degrees of freedom. In R notation, $\chi_{a,v}^2$ equals
 8 `qchisq(1-a, v)`.

1 *Confidence Interval for Population Proportion, p*

2 Assume that n is large enough that you are confident that $np \geq 5$ and
 3 $n(1 - p) \geq 5$. Let $\hat{p} = X/n$ where X is binomial(n, p), and p is unknown.
 4 Then, a $100(1 - \alpha)\%$ confidence interval for p is

$$5 \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

6 where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal
 7 distribution. In R notation, z_a equals `qnorm(1-a)`.

1 **Exercise:** A sample of size 50 is drawn from a population with unknown
2 mean and variance. It holds that $\bar{x} = 14.6$ and $s = 1.3$. Estimate the
3 population mean. State both the standard error for the estimator and a
4 90% confidence interval for μ .

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